# Lab4
# นาย ภูริณัฐ บุญมาถึง
# Packet Tracer - Investigate STP Loop Prevention

## Objectives

In this lab, you will observe spanning-tree port states and watch the spanning-tree convergence process. = Describe the operation of Spanning Tree Protocol.

= Explain how Spanning Tree Protocol prevents switching loops while allowing redundancy in switched networks. **Background / Scenario**

In this activity you will use Packet Tracer to observe the operation of Spanning Tree Protocol in a simple switched network that has redundant paths.

## Instructions

### Part 1: Observe a Converged Spanning-Tree Instance

#### Step 1: Verify Connectivity.

Ping from PC1 to PC2 to verify connectivity between the hosts. Your ping should be successful.

#### Step 2: View spanning-tree status on each switch.

Use the **show spanning-tree vlan 1** command to gather information about the spanning tree status of each switch. Complete the table. For the purposes of the activity, only consider information about the Gigabit trunk ports. The Fast Ethernet ports are access ports that have end devices connected and are not part of the inter-switch trunk-based spanning tree.

| Switch | Port | Status (FWD, BLKâ€¦) | Root Bridge? |
|---|---|---|---|
| S1 | G0/1 | | |
| | G0/2 | | |
| S2 | G0/1 | | |
| | G0/2 | | |
| S3 | G0/1 | | |
| | G0/2 | | |

Packet Tracer uses a different link light on one of the connections between the switches.

What do you think this this link light means?

What path will frames take from PC1 to PC2?

Why do the frames not travel through S3?

Why has spanning tree placed a port in blocking state?

## Part 2: Observe spanning-tree convergence

### Step 1: Remove the connection between S1 and S2.

a. Open a CLI window on switch S3 and issue the command **show spanning-tree vlan 1**. Leave the CLI window open.

b. Select the delete tool from the menu bar and click the cable that connects S1 and S2.

### Step 2: Observe spanning-tree convergence.

a. Quickly return to the CLI prompt on switch S3 and issue the **show spanning-tree vlan 1** command.

b. Use the up-arrow key to recall the **show spanning-tree vlan 1** command and issue it repeatedly until the orange link light on the cable turns green. Observe the status of port G0/2.

What do you see happen to the status of the G0/2 port during this process?

You have observed the transition in port status that occurs as a spanning-tree port moves from blocking to forwarding state. c. Verify Connectivity by pinging from PC1 to PC2. Your ping should be successful.

Are any ports showing an orange link light that indicates that the port is in a spanning-tree state other than forwarding? Why or why not?

ตอบ

# สรุปคำถาม-คำตอบ: การทำงานของ Spanning Tree Protocol (STP)

## ส่วนที่ 1: การทำงานของพอร์ตและการป้องกันการเกิดลูป

**คำถาม: คุณคิดว่าไฟสัญญาณเชื่อมต่อ (Link Light) สีส้ม/เหลืองนี้หมายถึงอะไร?**

> **คำตอบ:** ไฟสัญญาณสีส้มหมายความว่าพอร์ตนั้นอยู่ในสถานะ **Blocking (บล็อก)** พอร์ตนี้จะไม่ส่งต่อเฟรมข้อมูลใด ๆ เพื่อป้องกันไม่ให้เกิดลูป (Switching Loop) ในเครือข่าย

**คำถาม: เฟรมข้อมูลจะใช้เส้นทางใดในการเดินทางจาก PC1 ไปยัง PC2?**

> **คำตอบ:** ข้อมูลจะเดินทางผ่านเส้นทาง **PC1 → S1 → S2 → PC2** (โดยระบบจะข้ามลิงก์ที่ถูกบล็อกระหว่าง S1 และ S3 ไปเลย)

**คำถาม: ทำไมเฟรมข้อมูลถึงไม่เดินทางผ่านสวิตช์ S3?**

> **คำตอบ:** เพราะโปรโตคอล Spanning Tree (STP) ได้สั่งให้พอร์ต G0/2 บนสวิตช์ S3 อยู่ในสถานะ **Blocking** ทำให้ไม่สามารถส่งผ่านข้อมูลใด ๆ บนลิงก์เส้นนั้นได้

**คำถาม: ทำไม Spanning Tree ถึงต้องสั่งให้พอร์ตใดพอร์ตหนึ่งอยู่ในสถานะบล็อก (Blocking)?**

**คำตอบ:** เพื่อป้องกันการเกิด **Switching Loop** (หรือ Broadcast Storm) ในเครือข่าย เนื่องจากหากปล่อยให้เส้นทางสำรอง (Redundant Paths) ทั้งหมดเปิดใช้งานพร้อมกัน เฟรมข้อมูลจะวิ่งวนไปมาอย่างไม่มีที่สิ้นสุด ซึ่งจะส่งผลให้เครือข่ายล่มและทำให้ตาราง MAC Address ทำงานผิดพลาด

## ส่วนที่ 2: การเปลี่ยนแปลงสถานะและผลลัพธ์หลังตัดเส้นทางซ้ำซ้อน

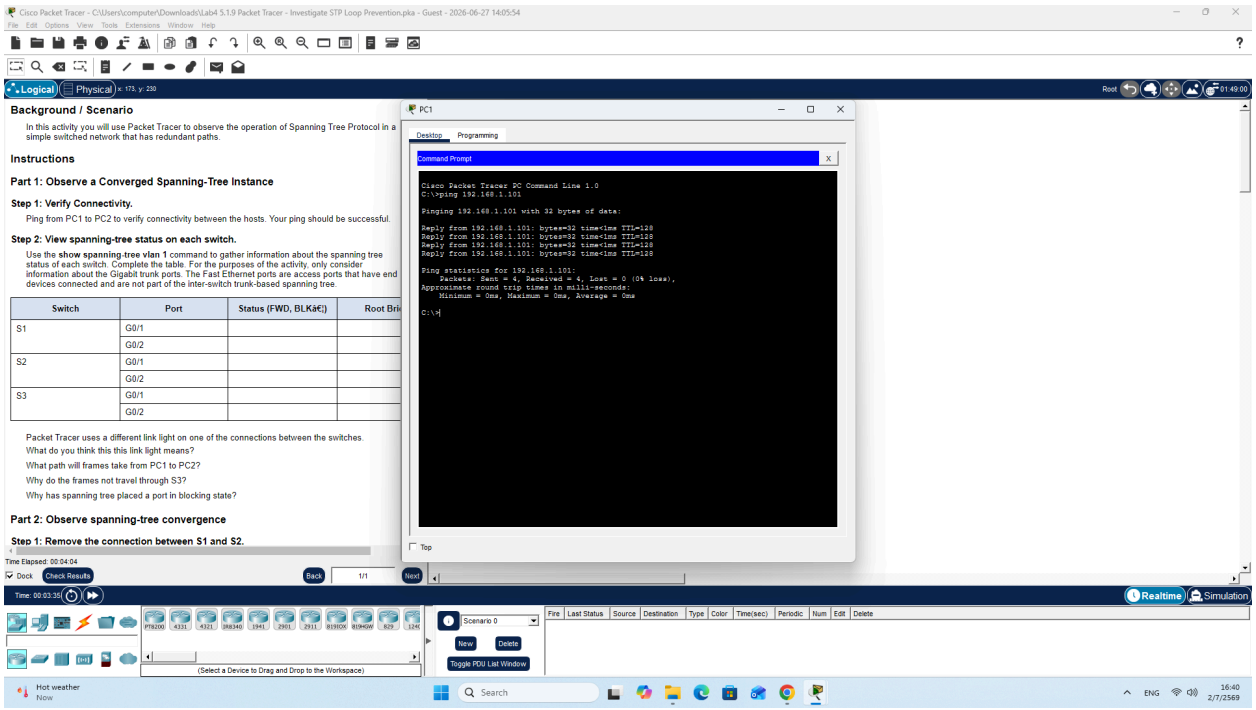
**คำถาม: คุณเห็นอะไรเกิดขึ้นกับสถานะของพอร์ต G0/2 ในระหว่างกระบวนการนี้?**

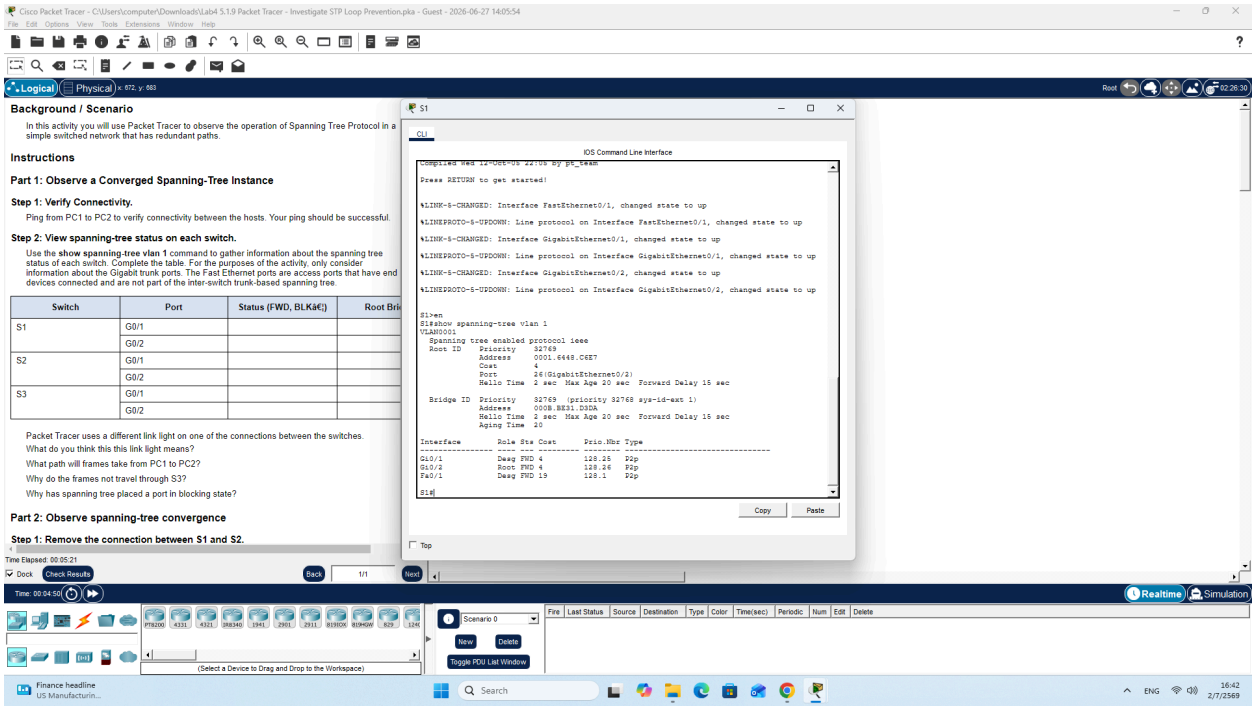
**คำตอบ:** พอร์ตจะค่อย ๆ เปลี่ยนสถานะตามมาตรฐานของ STP ตามลำดับดังนี้:

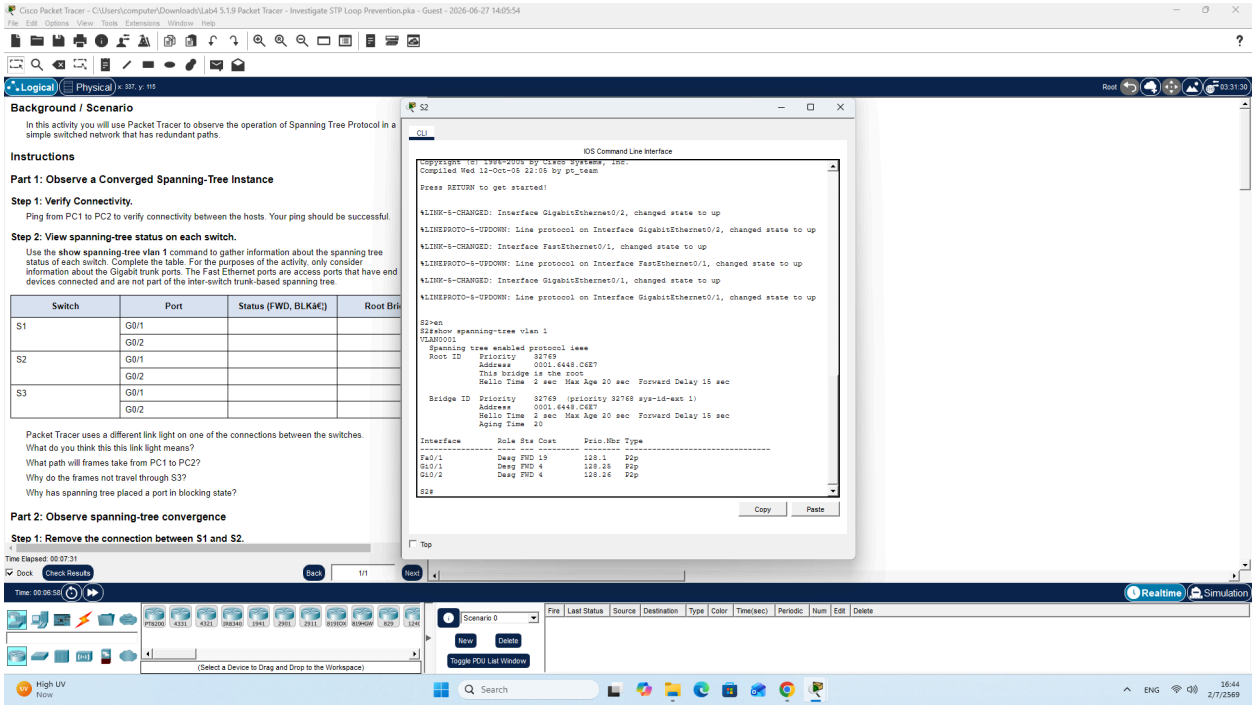
1. **BLK (Blocking):** สถานะบล็อก
2. **LRN (Learning):** พอร์ตกำลังเรียนรู้ MAC Address (ไฟสัญญาณยังคงเป็นสีส้ม)
3. **FWD (Forwarding):** สถานะพร้อมส่งต่อข้อมูลสำเร็จ (ไฟสัญญาณเปลี่ยนเป็นสีเขียว)

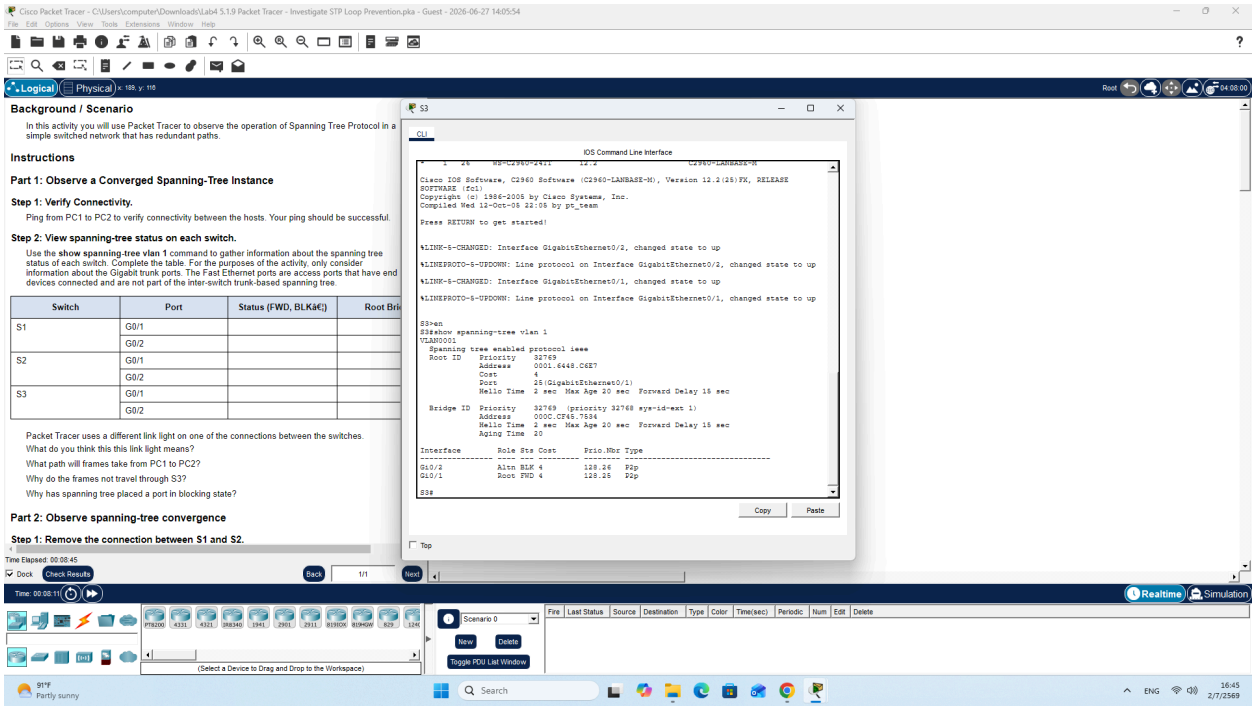
**คำถาม: มีพอร์ตใดที่ยังแสดงไฟสัญญาณสีส้ม (ที่ระบุว่าพอร์ตนั้นไม่ได้อยู่ในสถานะ Forwarding) อีกหรือไม่? เพราะเหตุใด?**

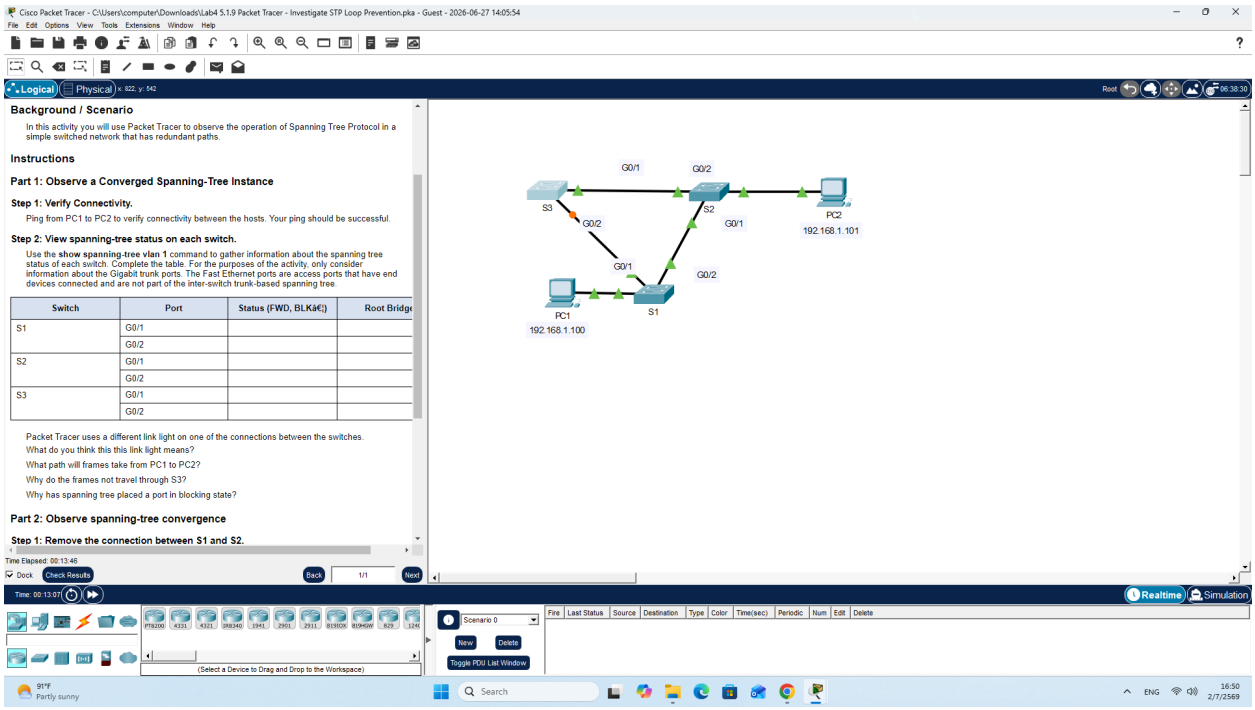
**คำตอบ: ไม่มีแล้ว** ไฟสัญญาณของพอร์ตทั้งหมดจะเปลี่ยนเป็นสีเขียวทั้งหมด **เหตุผล:** เนื่องจากในระบบไม่มีเส้นทางซ้ำซ้อน (Redundant Path) ที่จะทำให้เกิดลูปได้อีกต่อไป (หลังจากการจำลองตัดสายระหว่าง S1 และ S2 ออกไป) เมื่อไม่สามารถเกิดลูปในทางกายภาพได้ พอร์ตที่เหลืออยู่ทั้งหมดจึงต้องทำหน้าที่ส่งต่อข้อมูล (**Forwarding**) ตามปกติเพื่อรักษาการเชื่อมต่อของเครือข่ายเอาไว้

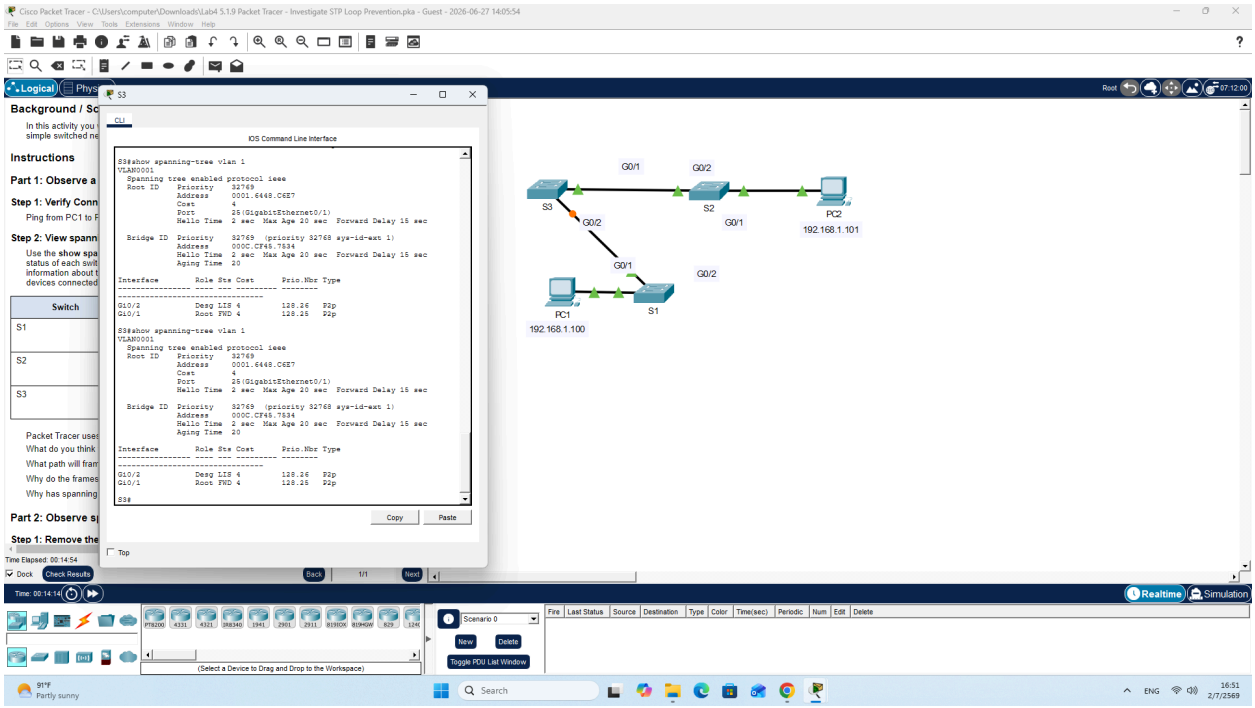


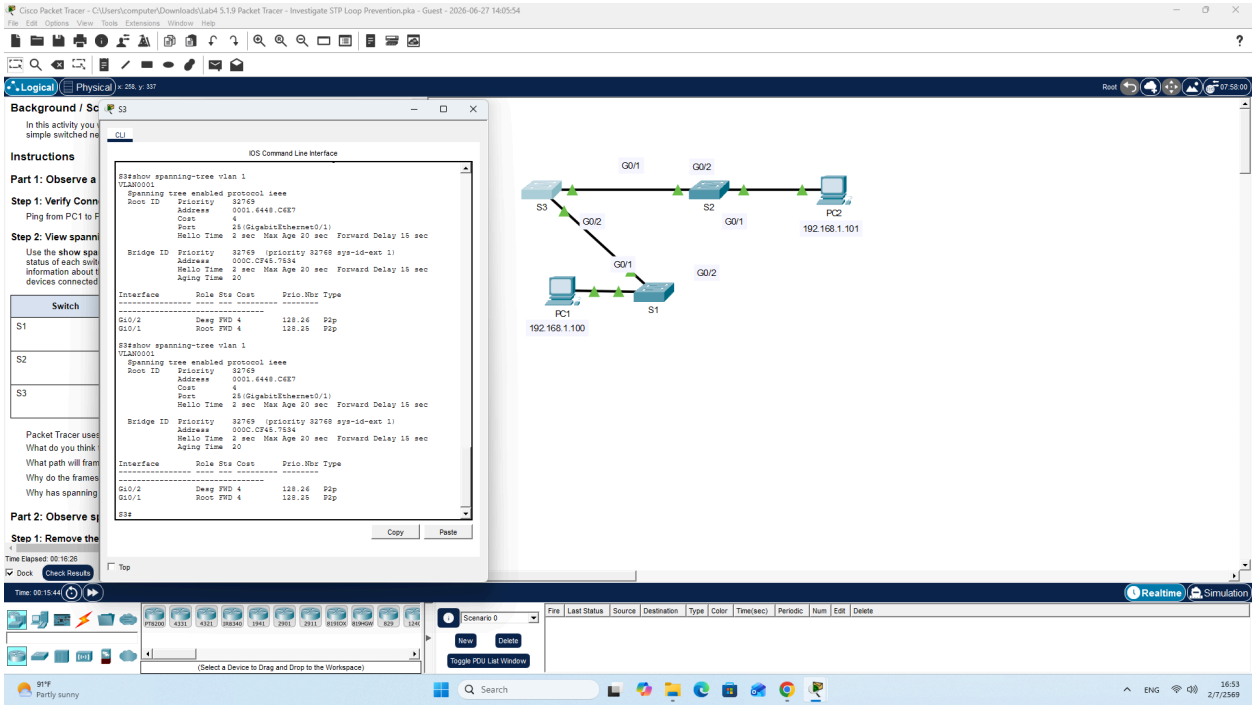

Background / Scenario
In this activity you will use Packet Tracer to observe the operation of Spanning Tree Protocol in a simple switched network that has redundant paths.
Instructions
Part 1: Observe a Converged Spanning-Tree Instance
Step 1: Verify Connectivity.
Ping from PC1 to PC2 to verify connectivity between the hosts. Your ping should be successful.
Step 2: View spanning-tree status on each switch.
Use the show spanning-tree vlan 1 command to gather information about the spanning tree status of each switch. Complete the table. For the purposes of the activity, only consider information about the Gigabit trunk ports. The Fast Ethernet ports are access ports that have end devices connected and are not part of the inter-switch trunk-based spanning tree.
Switch
Port
Status (FWD, BLK…)
S1
G0/1
G0/2
S2
G0/1
G0/2
S3
G0/1
G0/2
Packet Tracer uses a different link light on one of the connections between the switches.
What do you think this this link light means?
What path will frames take from PC1 to PC2?
Why do the frames not travel through S3?
Why has spanning tree placed a port in blocking state?
Part 2: Observe spanning-tree convergence
Step 1: Remove the connection between S1 and S2.
S1
IOS Command Line Interface
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
S1>en
S1#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 26(GigabitEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0008.A831.D05A
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/1 Desg FWD 4 128.25 P2p
Gi0/2 Root FWD 4 128.26 P2p
Fa0/1 Desg FWD 19 128.1 P2p
S1#
Copy
Paste
Top
Time Elapsed: 00:05:21
Check Results
Back
Next
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)

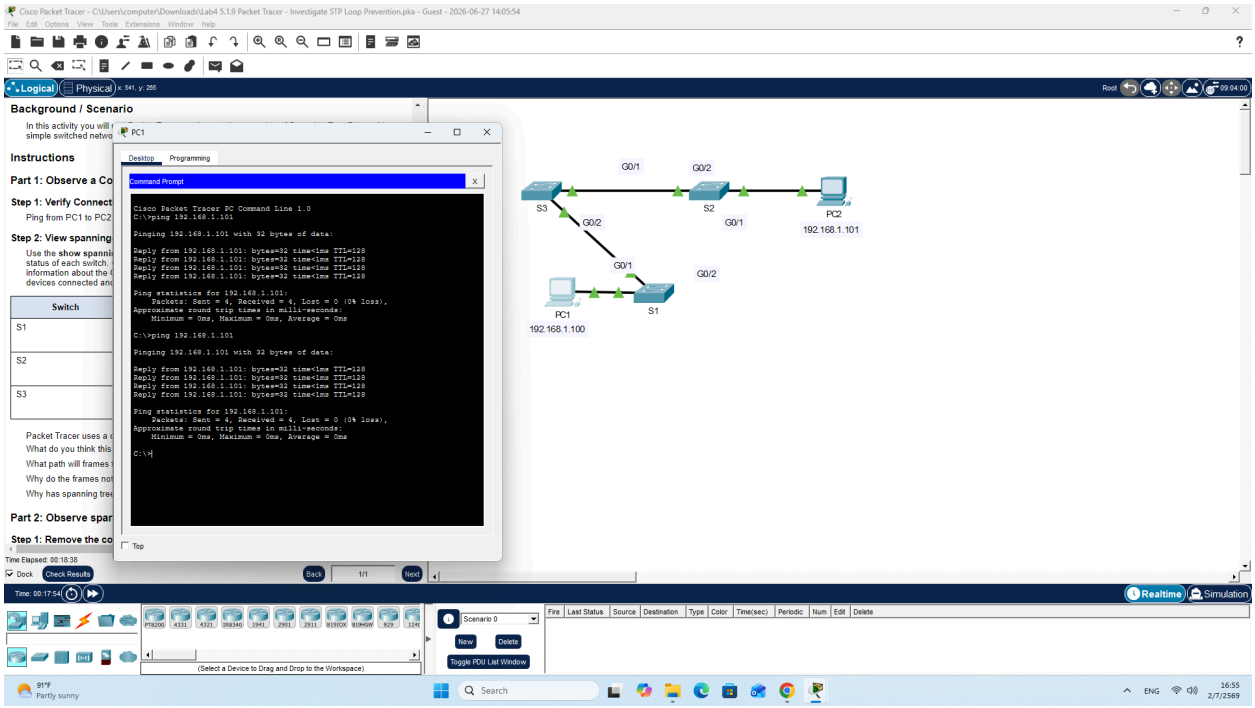

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Switch
Port
Status (FWD, BLK…)
S1
G0/1
G0/2
S2
G0/1
G0/2
S3
G0/1
G0/2
Packet Tracer uses a different link light on one of the connections between the switches.
What do you think this this link light means?
What path will frames take from PC1 to PC2?
Why do the frames not travel through S3?
Why has spanning tree placed a port in blocking state?
Part 2: Observe spanning-tree convergence
Step 1: Remove the connection between S1 and S2.
S2
IOS Command Line Interface
Compiled Wed 12-Oct-05 22:05 by pt_team
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
S2>en
S2#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0001.6448.C6E7
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Fa0/1 Desg FWD 19 128.1 P2p
Gi0/1 Desg FWD 4 128.25 P2p
Gi0/2 Desg FWD 4 128.26 P2p
S2#
Copy
Paste
Top
Time Elapsed: 00:07:31
Check Results
Back
Next
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)


Cisco Packet Tracer - C:\Users\computer\Downloads\Lab4 5.1.9 Packet Tracer - Investigate STP Loop Prevention.pka - Guest - 2026-06-27 14:05:54
Logical
Physical
Background / Scenario
In this activity you will use Packet Tracer to observe the operation of Spanning Tree Protocol in a simple switched network that has redundant paths.
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Switch
Port
Status (FWD, BLK…)
S1
G0/1
G0/2
S2
G0/1
G0/2
S3
G0/1
G0/2
Packet Tracer uses a different link light on one of the connections between the switches.
What do you think this this link light means?
What path will frames take from PC1 to PC2?
Why do the frames not travel through S3?
Why has spanning tree placed a port in blocking state?
Part 2: Observe spanning-tree convergence
Step 1: Remove the connection between S1 and S2.
Time Elapsed: 00:08:45
Check Results
Back
Next
S3
CLI
IOS Command Line Interface
Cisco IOS Software, C2960 Software (C2960-LANBASE-M), Version 12.2(25)FX, RELEASE SOFTWARE (fc1)
Copyright (c) 1986-2005 by Cisco Systems, Inc.
Compiled Wed 12-Oct-05 22:05 by pt_team
Press RETURN to get started!
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
S3>en
S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF45.7534
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/2 Altn BLK 4 128.26 P2p
Gi0/1 Root FWD 4 128.25 P2p
S3#
Copy
Paste
Top
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
91°F
Partly sunny
Search
ENG
16:45
2/7/2569


Cisco Packet Tracer - C:\Users\computer\Downloads\Lab4 5.1.9 Packet Tracer - Investigate STP Loop Prevention.pka - Guest - 2026-06-27 14:05:54
Logical
Physical
Background / Scenario
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Switch
Port
Status (FWD, BLK…)
S1
G0/1
G0/2
S2
G0/1
G0/2
S3
G0/1
G0/2
Packet Tracer uses a different link light on one of the connections between the switches.
What do you think this this link light means?
What path will frames take from PC1 to PC2?
Why do the frames not travel through S3?
Why has spanning tree placed a port in blocking state?
Part 2: Observe spanning-tree convergence
Step 1: Remove the connection between S1 and S2.
Time Elapsed: 00:13:46
Check Results
Back
Next
G0/1
G0/2
S3
S2
G0/2
G0/1
PC2
192.168.1.101
G0/1
G0/2
PC1
192.168.1.100
S1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
91°F
Partly sunny
Search
ENG
16:50
2/7/2569


S3
IOS Command Line Interface
S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF45.7534
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/2 Desg LIS 4 128.26 P2p
Gi0/1 Root FWD 4 128.25 P2p
S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF45.7534
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/2 Desg LIS 4 128.26 P2p
Gi0/1 Root FWD 4 128.25 P2p
S3#
Copy
Paste
Top
G0/1
G0/2
S3
S2
PC2
192.168.1.101
G0/2
G0/1
G0/1
G0/2
PC1
S1
192.168.1.100


S3
IOS Command Line Interface
S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF45.7534
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/2 Desg FWD 4 128.26 P2p
Gi0/1 Root FWD 4 128.25 P2p
S3#show spanning-tree vlan 1
VLAN0001
Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0001.6448.C6E7
Cost 4
Port 25(GigabitEthernet0/1)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 000C.CF45.7534
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20
Interface Role Sts Cost Prio.Nbr Type
Gi0/2 Desg FWD 4 128.26 P2p
Gi0/1 Root FWD 4 128.25 P2p
S3#
Copy
Paste
Top
G0/1
G0/2
S3
S2
PC2
192.168.1.101
G0/2
G0/1
G0/1
G0/2
PC1
S1
192.168.1.100


Cisco Packet Tracer - C:\Users\computer\Downloads\Lab4 5.1.9 Packet Tracer - Investigate STP Loop Prevention.pka - Guest - 2026-06-27 14:05:54
Logical
Physical
Background / Scenario
Instructions
Part 1: Observe a Co
Step 1: Verify Connect
Step 2: View spanning
Switch
S1
S2
S3
Part 2: Observe spar
Step 1: Remove the co
PC1
Desktop
Programming
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.101
Pinging 192.168.1.101 with 32 bytes of data:
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.101:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.1.101
Pinging 192.168.1.101 with 32 bytes of data:
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
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Reply from 192.168.1.101: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.101:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top
Back
Next
G0/1
G0/2
S3
S2
PC2
192.168.1.101
PC1
192.168.1.100
S1
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
(Select a Device to Drag and Drop to the Workspace)
Fire
Last Status
Source
Destination
Type
Color
Time(sec)
Periodic
Num
Edit
Delete
Search
ENG
16:55
2/7/2569